CS285 HW1 — MSE Action-Chunking Policy Report

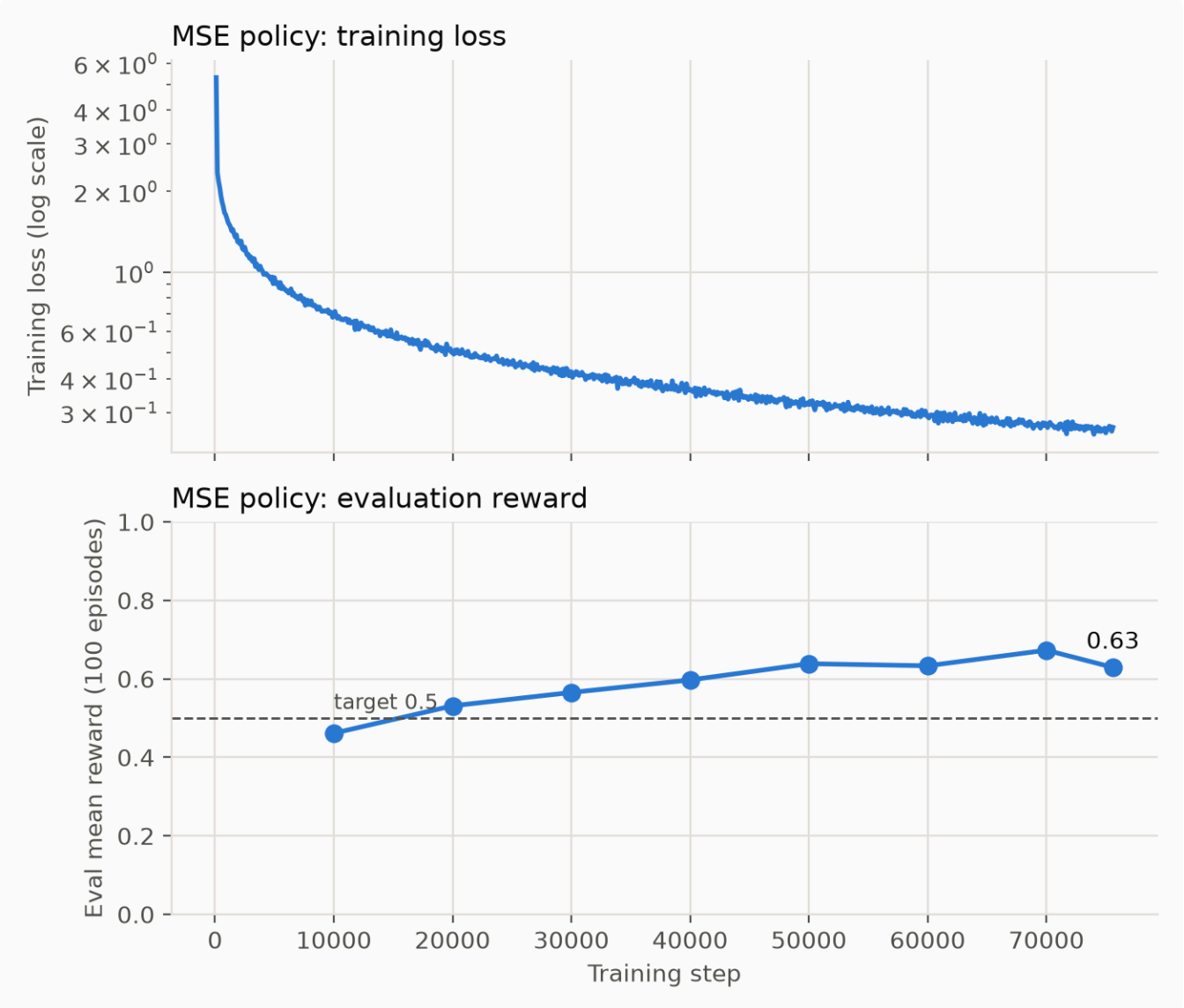
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Result

The MSE action-chunking policy reaches a final evaluation reward of **0.629** on Push-T, above the 0.5 target. Its best checkpoint scored **0.673** at step 70,000.

Reward is the mean, over 100 evaluation episodes, of each episode's maximum reward. The full run (75,600 steps) took 6 min 52 s on 8 CPU cores. Run logs and rollout videos: [WandB run 2lv8utm3](2lv8utm3).

Training curves



Training loss falls from 5.30 at step 100 to 0.26 at the end and is still slowly decreasing. Evaluation reward crosses the 0.5 target by step 20,000 and levels off around 0.63–0.67 after step 50,000, so lower training loss stops buying reward. Plotted from the run's logged metrics with matplotlib (`plot_curves.py`), not from WandB screenshots.

MLP architecture

The policy is a 4-layer fully connected MLP: 3 hidden layers of 256 units with ReLU, then a linear output layer. It maps the 5-D normalized state to a whole action chunk in one forward pass. The 16 outputs are reshaped to 8 actions × 2 dimensions (chunk size K = 8).

Layer	Input → output	Activation	Parameters
Hidden 1	5 → 256	ReLU	1,536
Hidden 2	256 → 256	ReLU	65,792
Hidden 3	256 → 256	ReLU	65,792
Output	256 → 16 (reshaped to 8 × 2)	none (linear)	4,112
Total			**137,232**

The output layer has no activation, because the targets are normalized actions that can be negative. There is no dropout, normalization layer or weight decay.

Training setup

The loss is Eq. 1: the squared L2 error over the whole predicted chunk, averaged over the batch.

```
L_{MSE}(\theta) = \frac{1}{B} \sum_{j=1}^{B} \left\| A_t^{(j)} -
\pi_\theta(o_t^{(j)}) \right\|_2^2
```

Setting	Value
Data	Push-T, 206 expert episodes, 25,650 transitions → 24,208 (state, 8-action chunk) windows
Normalization	per-feature z-score of states and actions (dataset mean and std)
Optimizer	Adam, learning rate 3e-4, no weight decay

Setting	Value
Batch size	128
Training length	400 epochs = 75,600 steps
Speed-up	`torch.compile` on the loss and its backward pass
Evaluation	every 10,000 steps and at the end: 100 episodes, chunks run open-loop, 5 rollout videos to WandB
Seed	42

All hyperparameters are the starter-code defaults; no sweep was run.